



2.1 Defining Average and Instantaneous Rates of Change at a Point

2.2 Defining the Derivative of a Function and Using Derivative Notation

Ex 1: Find the slope of...

a) $y = 5x + 4$

$$m = 5$$

b) $y - 3 = -2(x + 1)$

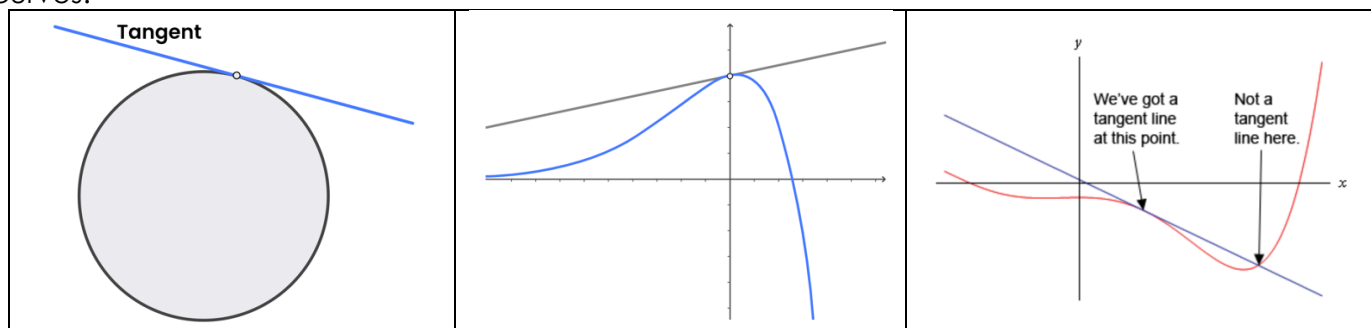
$$m = -2$$

c) The line that passes through the points (0, 3) and (6, -7).

$$m = -\frac{5}{3}$$

What is a tangent line?

A tangent line, or simply tangent, is a line that touches a graph at one point in a small neighborhood around the point of tangency. Tangent lines can be drawn at points on circles or curves.



These lines may be tangent to a curve at one point but can later cross the curve again at a point far from the point of tangency.

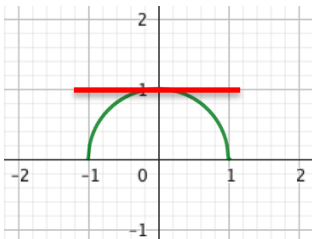
At the point of tangency the curve and the line share two important properties:

1. **point**

2. **slope**

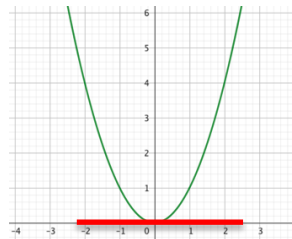
Ex 2: Draw a tangent line at the indicated point. Then, find the slope of the tangent line (and thus the curve) at the indicated point.

a) (0, 1)



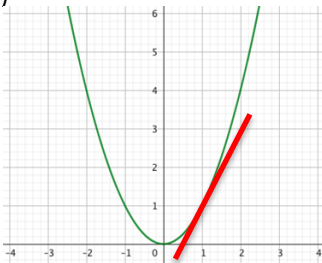
$$m = 0$$

b) (0, 0)



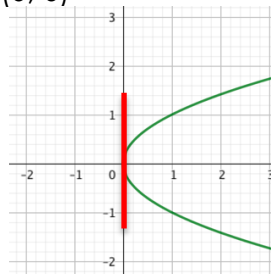
$$m = 0$$

c) (1, 1)



$$m = 2$$

d) (0, 0)



m is undefined

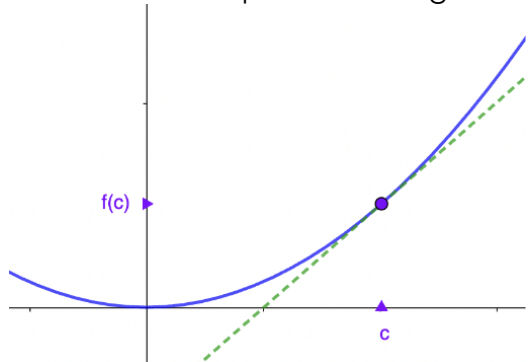


The Tangent Line Problem

Calculus grew out of four major problems that European mathematicians were working on in the seventeenth century. The first of the four problems is known as the "Tangent Line Problem."



Task: Find the slope of the tangent line to $f(x)$ at $x=c$.



$$m_{sec} = \frac{f(c+h) - f(c)}{c+h-c}$$

$$m_{tan} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

Definition of Tangent Line with Slope m

If f is defined on an open interval containing c , and if the limit

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

exists, then the line passing through $(c, f(c))$ with slope m is the **tangent line** to the graph of f at the point $(c, f(c))$.

Ex 3: Find the slope of f at the given point.

a) $f(x) = x^2 + 1, \quad (2, 5)$

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ m &= \lim_{h \rightarrow 0} \frac{(2+h)^2 + 1 - 5}{h} \\ m &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 + 1 - 5}{h} \\ m &= \lim_{h \rightarrow 0} \frac{h(4+h)}{h} \\ m &= 4 \end{aligned}$$

b) $f(x) = x^2 + 1, \quad (-1, 2)$

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} \\ m &= \lim_{h \rightarrow 0} \frac{(-1+h)^2 + 1 - 2}{h} \\ m &= \lim_{h \rightarrow 0} \frac{1 - 2h + h^2 + 1 - 2}{h} \\ m &= \lim_{h \rightarrow 0} \frac{h(-2+h)}{h} \\ m &= -2 \end{aligned}$$

c) $f(x) = x^2 + 1, \quad (0, 1)$

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ m &= \lim_{h \rightarrow 0} \frac{(0+h)^2 + 1 - 1}{h} \\ m &= \lim_{h \rightarrow 0} \frac{h^2}{h} \\ m &= \lim_{h \rightarrow 0} h \\ m &= 0 \end{aligned}$$

d) $f(x) = 3, \quad (21, 3)$

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(21+h) - f(21)}{h} \\ m &= \lim_{h \rightarrow 0} \frac{3 - 3}{h} \\ m &= \lim_{h \rightarrow 0} \frac{0}{h} \\ m &= \lim_{h \rightarrow 0} 0 \\ m &= 0 \end{aligned}$$

Ex 4: Interpret the expression: $\lim_{h \rightarrow 0} \frac{g(6+h) - g(6)}{h} = -5$

The slope of g at $x = 6$ is -5 .



The Derivative of a Function

The **derivative** of f at x is

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

provided the limit exists. For all x for which this limit exists, f' is a function of x .

The notation $f'(x)$ is read f prime of x .

What is a Derivative?

A derivative is a function used to calculate slope.

The derivative can also be used to determine the **instantaneous** rate of change (or simply the rate of change) of one variable with respect to another.

Vocabulary

- **differentiation** - the process of finding a derivative
- **differentiate** - to find a derivative
- **differentiable** - a derivative exists



"derive" and
"differentiate" are
NOT synonyms!!

Notation

There are three different notations we will use to describe derivatives.

	Type	Notation	Example
1	Lagrange	$f'(x)$	If $f(x) = x^2$, then $f'(x) = 2x$.
2	Leibniz	$\frac{dy}{dx}$	If $y = x^2$, then $\frac{dy}{dx} = 2x$.
3	Leibniz – verb form	$\frac{d}{dx}[f(x)]$	$\frac{d}{dx}[x^2] = 2x$

Ex 5: Find the derivative using the limit process.

a) $f(x) = x^2$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$ $f'(x) = 2x$	b) $g(x) = \sqrt{x}$ $g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$ $g'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$ $g'(x) = \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})}$ $g'(x) = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$ $g'(x) = \frac{1}{2\sqrt{x}}$	c) $y = \frac{1}{x}$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \cdot \frac{x(x+h)}{x(x+h)}$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{x - (x+h)}{xh(x+h)}$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{-h}{xh(x+h)}$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)}$ $\frac{dy}{dx} = -\frac{1}{x^2}$
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Ex 6: Find the slope at the given point.

a) $f(x) = x^2$, $x = -2$ $f'(-2) = -4$	b) $g(x) = \sqrt{x}$, $x = 9$ $g'(9) = \frac{1}{6}$	c) $y = \frac{1}{x}$, $x = 10$ $\left. \frac{dy}{dx} \right _{x=10} = -\frac{1}{100}$
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Ex 7: Interpret the expression: $h'(20) = 25$

The slope of h at $x = 20$ is 25.



Alternate Form

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$



Alternate form does NOT yield derivatives, it yields slopes!

Ex 8: Use the alternate form to find the slope at the indicated point.

a) $f(x) = x^2 + 1, (2, 5)$

$$f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$$

$$f'(2) = \lim_{x \rightarrow 2} \frac{x^2 + 1 - 5}{x - 2}$$

$$f'(2) = \lim_{x \rightarrow 2} \frac{(x+2)(\cancel{x-2})}{\cancel{x-2}}$$

$$f'(2) = 4$$

b) $g(x) = 17 - 4x, (1, 13)$

$$g'(1) = \lim_{x \rightarrow 1} \frac{g(x) - g(1)}{x - 1}$$

$$g'(1) = \lim_{x \rightarrow 1} \frac{17 - 4x - 13}{x - 1}$$

$$g'(1) = \lim_{x \rightarrow 1} \frac{-4(\cancel{x-1})}{\cancel{x-1}}$$

$$g'(1) = -4$$

c) $h(x) = |x|, (0, 0)$

$$h'(0) = \lim_{x \rightarrow 0} \frac{h(x) - h(0)}{x - 0}$$

$$h'(0) = \lim_{x \rightarrow 0} \frac{|x| - 0}{x - 0}$$

$$h'(0)^- = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$h'(0)^+ = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

$$h'(0) \text{ DNE}$$

Ex 9:

What is $\lim_{h \rightarrow 0} \frac{\cos(\pi + h) - \cos(\pi)}{h}$?

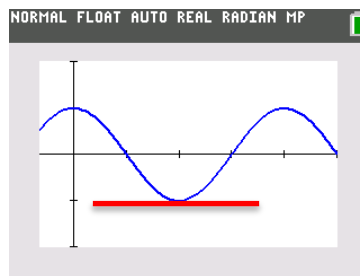
(A) 1

(B) $\frac{\sqrt{2}}{2}$

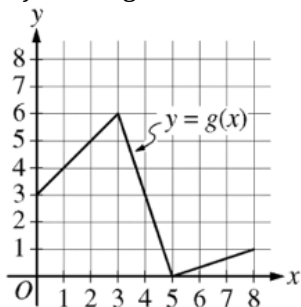
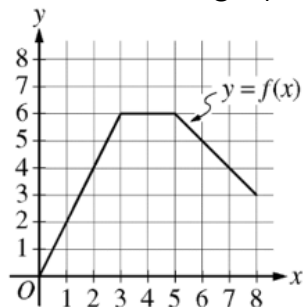
(C) 0

(D) -1

(E) The limit does not exist.



Ex 10: Use the graphs of f and g below to evaluate each expression.



a) $\lim_{x \rightarrow 4} \frac{g(x) - g(4)}{x - 4}$

$$\lim_{x \rightarrow 4} \frac{g(x) - g(4)}{x - 4} = g'(4)$$

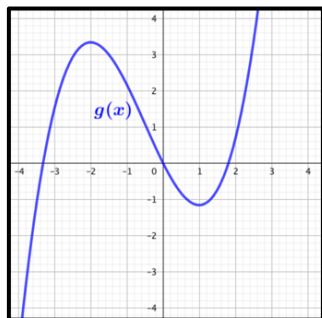
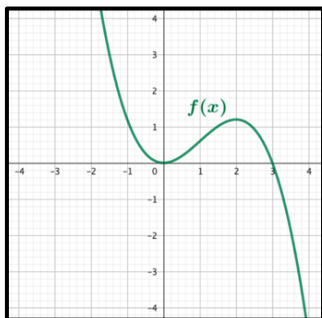
$$g'(4) = -3$$

b) $\lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$

$$\lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h} = f'(4)$$

$$f'(4) = 0$$

Ex 11: Use the graphs of f and g to arrange the values from least to greatest:



I. $g'(-1)$

II. $f'(1)$

III. $f'(0)$

$$\text{I} < \text{III} < \text{II}$$